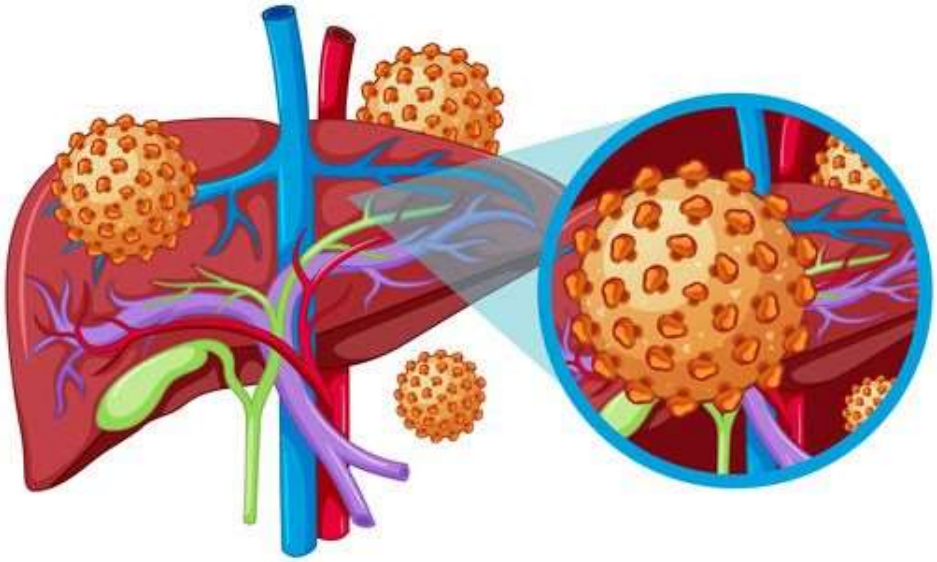
HEPATITIS



Dr. Taneem M. Ali

Viral Hepatitis Overview

Hepatitis B and C are major global health problems

Both affect the liver and may lead to chronic disease

Complications include cirrhosis and liver cancer

Problem Statement

- Hepatitis B is endemic in almost all part of world
- 60% of world population live in endemic area
- Estimated 2 billion people infected
- 360 million live with chronic infection
- 600,000 person dies annually as result of consequence of hepatitis B every year
- Estimated 25% child dies in later life as a consequence of hepatitis B infection

Hepatitis B - Problem Statement

257million people live with chronic HBV

Causes ~887,000 deaths annually

Major burden in Africa and Western Pacific

HBV - Epidemiology

Transmission: Blood and body fluids

Reservoir: Humans

Agent: DNA virus (Hepadnavirus)

High infectivity in HBeAg positive cases

HBV - Transmission

- Parenteral (needles, transfusion)
- Perinatal (mother to child)
- Sexual transmission
- Contaminated instruments

Risk factors includes:

- 1) Age
- the outcome of infection is age-dependent. The development of chronic hepatitis B infection is inversely related to age.
- 2) High risk groups
- occupational (physicians, health care and laboratory personnel)
- blood recipients and recipients of solid organ transplants
- Homosexuals
- percutaneous drug abusers,
- Immune compromised patients
- 3) Hepatitis B and HIV infection:
- 1% of HIV infected persons are infected with hepatitis B virus

HBV - Diagnosis

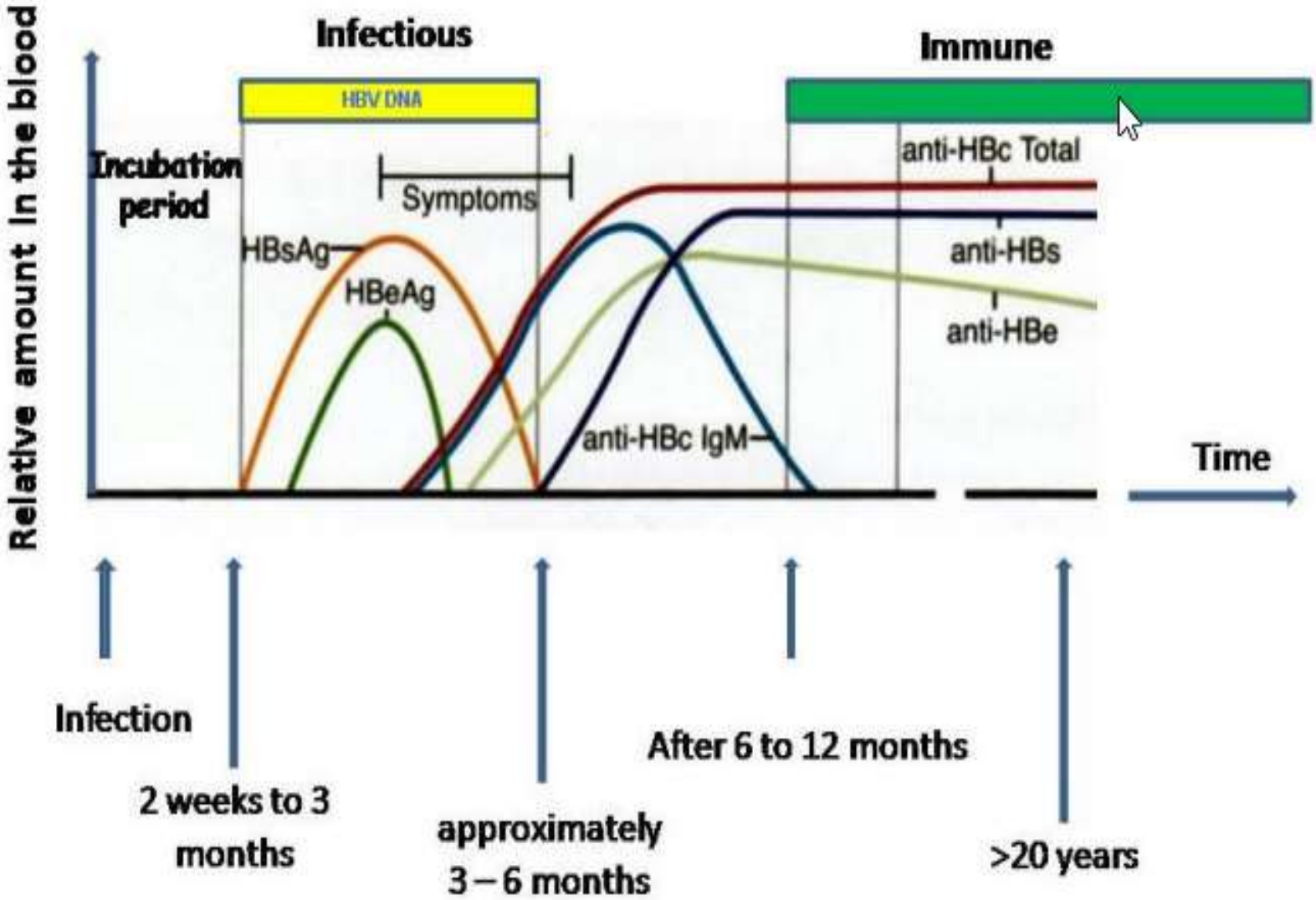
HBsAg: indicates infection

Anti-HBs: immunity

HBeAg: high infectivity

HBV DNA: confirms replication

HBV antigens and antibodies in the blood



HBS Ag

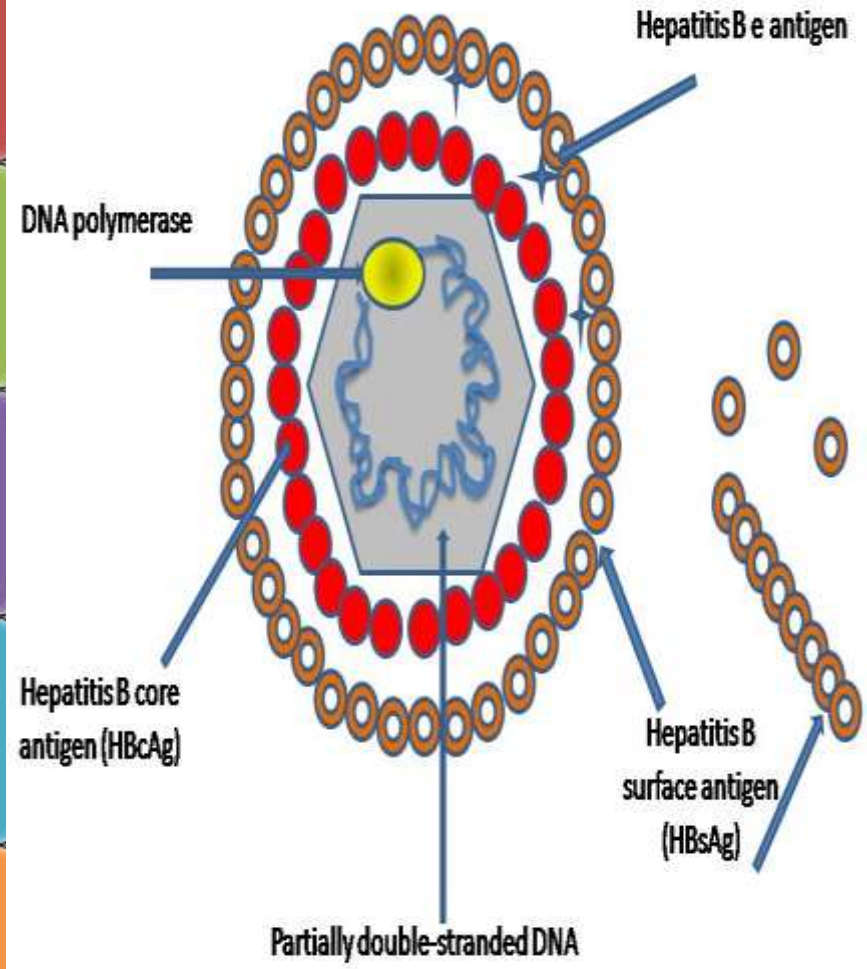
It is the first marker identified.

It was found in Blood as spherical or tubular form (18-22nm).

It is the protein which makes the virus coat.

It is detectable in **early phase** following infection (viraemia) or in **chronic carrier state**.

Chronic carrier is the person whose serum is repeatedly HBs Ag positive, over a period of 6 months or longer.



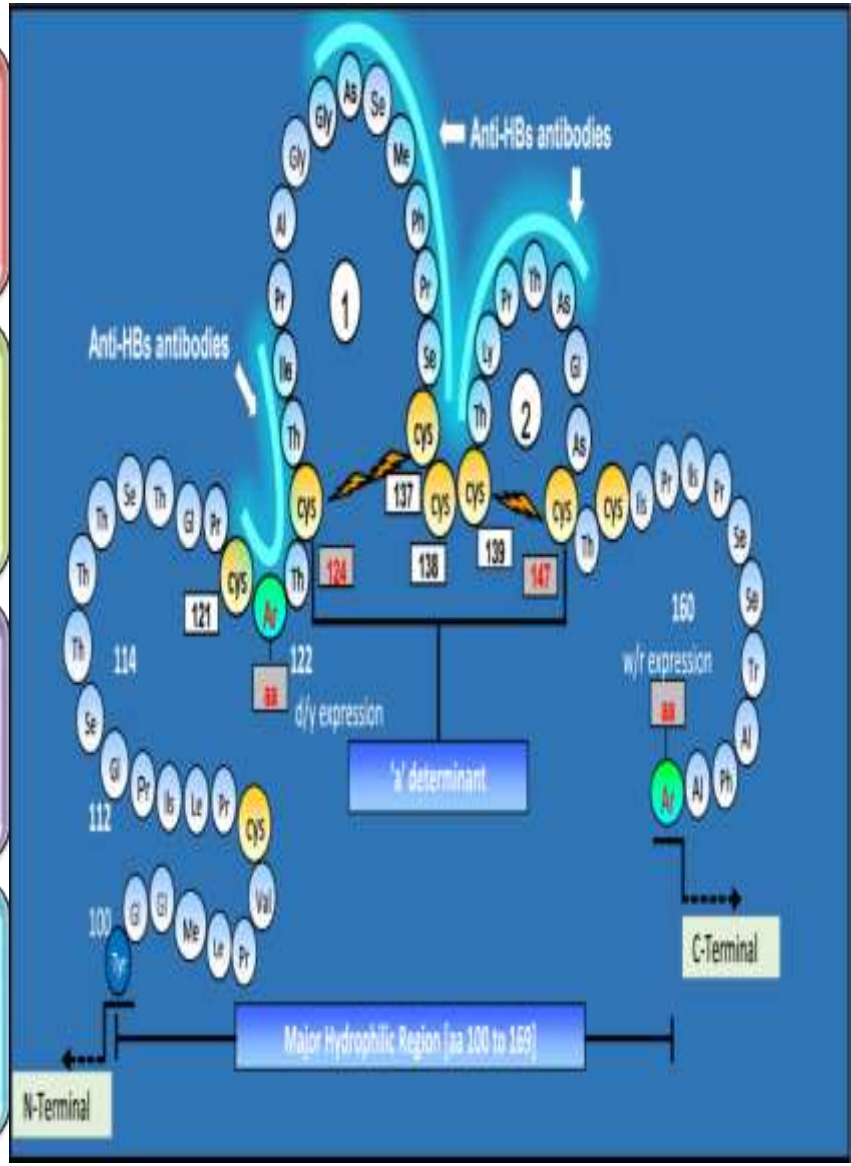
Anti-HBs:

antibodies against surface antigen

If HBs Ag disappears, it is usually followed after few months by the appearance of antiHBS antibodies which often remain for life.

It is the last antibody to appear

Presence of anti HBS indicates immunity



anti-HBc

antibodies against core Ag

It is the first antibody to appear.

It is not protective and plays no role in the immune pathogenesis.

It indicates previous infection by HB virus (either the person is still diseased or became cured and immune)

1-HBe Ag. its presence indicates:

The virus is actively replicated.

The patient is highly infectious

The patient is more liable to complications

2- **Anti-HBe**: (Antibodies Against e antigen) its presence medicates the virus is slowly

replicated, the patient is less infectious the patient is less liable to complication.

HBV - Prevention

- Vaccination (6 ,1 ,0months)
- Safe injections and blood screening
- Target high-risk groups
- HBIG for post-exposure

Control of patient and contact

- Immunization of contacts:
  - HBIG and hepatitis vaccine

a) Infants born to HBsAg positive mothers should be given a single dose of HBIG (0.5ml IM) and vaccine within 12 hours of birth. The first dose of vaccine should be given concurrently with HBIG at birth.

b) After percutaneous (e.g., needle stick) or mucous membrane exposures to blood that contains or might contain HbsAg.

c) After sexual exposure to a person with acute HBV infection, a single dose of HBIG (0.06 ml/kg) is recommended.

Control of patient and contact

- Specific treatment: No specific treatment is available for acute hepatitis B. Alpha
- interferon and lamivudine have been licensed for treatment of chronic hepatitis B in
- the USA. Studies have shown that alpha interferon is successful in arresting viral
- replication in about 25%-40% of treated" patients.
- VIRAL HEPAIITI

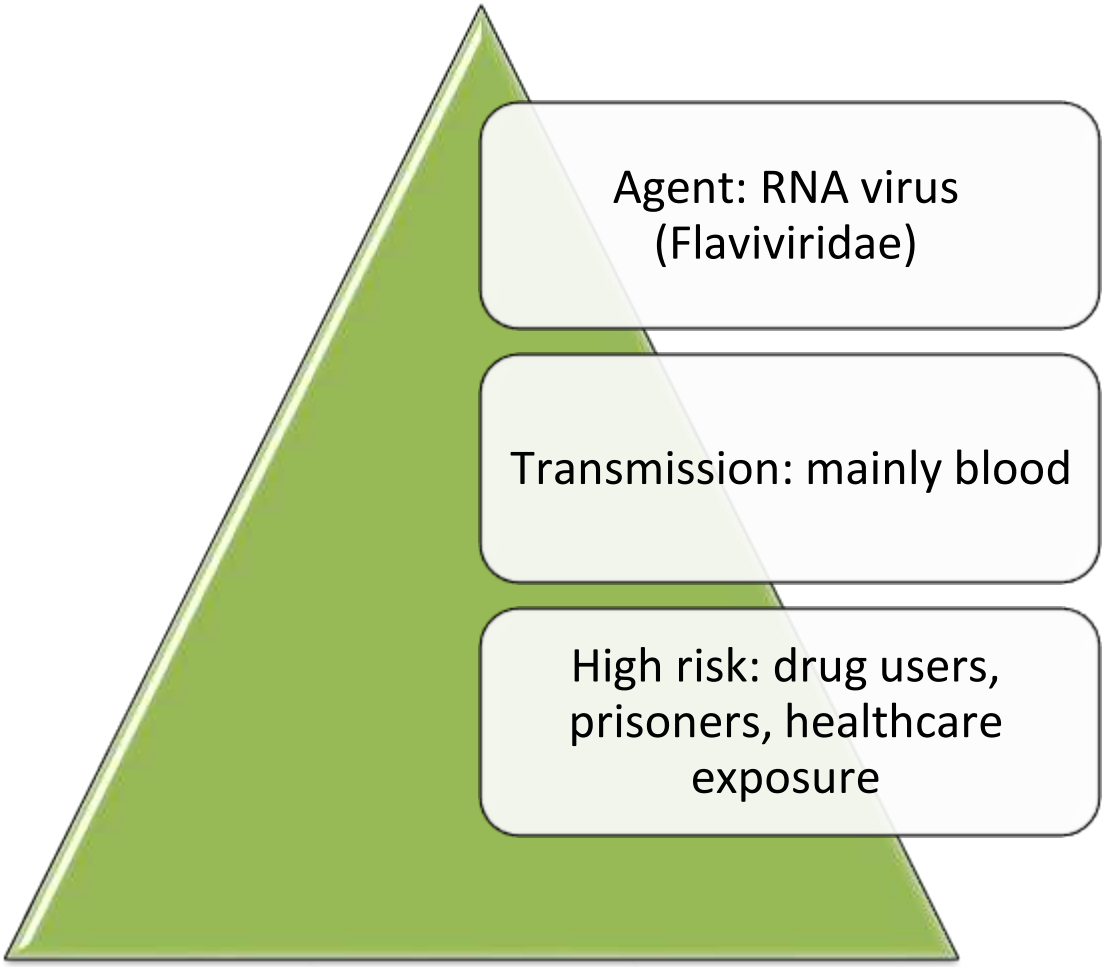
Hepatitis C - Problem Statement

71million people chronically infected

Major cause of liver cancer

No vaccine available

HCV - Epidemiology



HCV - Diagnosis

- Anti-HCV antibodies screening
- HCV RNA confirms infection
- Often asymptomatic early

HCV - Prevention

- Safe injections
- Screening blood products
- Harm reduction for drug users
- Infection control in healthcare

HCV - Management

- Antiviral therapy cures >95%
- Regular monitoring
- Vaccinate against HBV & HAV
- Education and counseling